LAB # 4

Yusuf Bozdağ 33840 Lab Section : C

1. Introduction

Operational amplifiers (op-amps) are fundamental building blocks in analog electronics and are widely used in signal amplification, filtering, and signal processing applications. An operational amplifier is a high-gain differential amplifier that amplifies the voltage difference between its input terminals while ideally drawing negligible input current. By applying appropriate feedback networks using resistors, the behavior and gain of an op-amp circuit can be precisely controlled, making feedback an essential concept in practical analog circuit design.

The objective of this experiment is to study the operation and characteristics of operational amplifier circuits, specifically inverting and non-inverting amplifier configurations. In this laboratory work, op-amp circuits are implemented using devices such as the LM741 or TL081C and analyzed through both DC and AC measurements. The experiment focuses on how feedback networks determine the voltage gain of the amplifier and how circuit parameters influence input and output behavior.

In the first part of the experiment, an inverting amplifier configuration is constructed and analyzed. In this configuration, the input signal is applied to the inverting terminal through a resistor while a feedback resistor connects the output to the same node. Due to the high open-loop gain of the op-amp and negative feedback, the inverting input behaves as a virtual ground. This property allows the voltage gain of the circuit to be determined primarily by the ratio of the feedback resistor to the input resistor. Measurements of currents, voltage gain, and input impedance are used to verify the theoretical predictions of the inverting amplifier model.

The dynamic behavior of the operational amplifier is also investigated using sinusoidal and square wave inputs. Using an oscilloscope, the relationship between input and output signals is observed, including the phase inversion characteristic of the inverting amplifier. In addition, rise time, fall time, and slew rate are analyzed to evaluate how quickly the op-amp can respond to rapidly changing input signals.

In the second part of the experiment, a non-inverting amplifier configuration is studied. In this configuration, the input signal is applied directly to the non-inverting terminal, resulting in an output signal that is in phase with the input. The voltage gain is determined by the feedback resistor network connected between the output and the inverting terminal. Experimental measurements are used to confirm the theoretical behavior of the non-inverting amplifier and to evaluate its input impedance characteristics.

Introduction

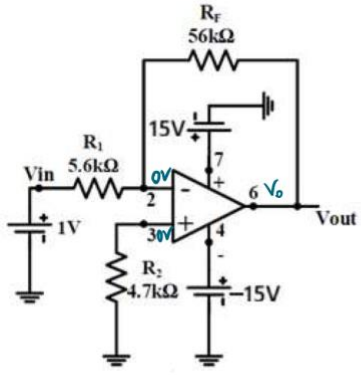
This pre-lab report presents the theoretical analysis of two fundamental op-amp configurations: the inverting amplifier and the non-inverting amplifier. The circuits will be implemented in the lab using the LM741 op-amp. The transfer functions (Vout/Vin) are derived using ideal op-amp assumptions.

The two ideal op-amp assumptions used throughout this analysis are:

1. The input impedance is infinite, meaning no current flows into the input terminals ($I_1 = I_2 = 0$).

2. The voltages at the inverting (−) and non-inverting (+) input terminals are equal (virtual short condition).

Part 1 – Inverting Amplifier



$$\frac{0 - V_{in}}{5.6} = \frac{V_o - 0}{56}$$

$$-56\, V_{in} = 56\, V_o$$

$$\frac{V_o}{V_{in}} = -10$$

Figure 1

Circuit Description

The inverting amplifier in Figure 1 consists of:

- R_1 = 5.6 kΩ (input resistor)
- R_F = 56 kΩ (feedback resistor)
- R_2 = 4.7 kΩ (connected from non-inverting input to ground)
- V_in = 1 V DC, ±15 V supply

Transfer Function Derivation

Since the non-inverting input (+) is connected to ground through R_2, and using the virtual short condition:

$$V^+ = 0 \Rightarrow V^- = V^+ = 0$$

Applying KCL at the inverting input node (pin 2), since no current enters the op-amp:

$$I_R1 = I_RF$$

Writing the currents in terms of voltages:

$$I_R1 = (V_in - 0) / R_1 = V_in / R_1$$

$$I_\{RF\} = (0 - V_out) / R_F = -V_out / R_F$$

Setting them equal:

$$V_in / R_1 = -V_out / R_F$$

Solving for the voltage gain:

$$A_v = V_out / V_in = -R_F / R_1$$

Numerical Calculation

$$A_v = -56\ k\Omega / 5.6\ k\Omega = -10$$

Expected output: V_out = −10 × 1 V = −10 V

The negative sign confirms that the output is 180° out of phase with the input.

Current Through Each Resistor (Expected Values)

I_R1 = V_in / R_1 = 1 V / 5.6 kΩ ≈ 0.179 mA

I_RF = |V_out| / R_F = 10 V / 56 kΩ ≈ 0.179 mA

I_R2 = V^+ / R_2 = 0 / 4.7 kΩ = 0 mA (pin 3 is at ground)

The fact that I_3 = 0 and V_pin2 ≈ 0 V confirms the virtual ground condition at pin 2.

Input Impedance

For the inverting amplifier, the input impedance is determined by the input resistor:

$$R_in = R_1 = 5.6\ k\Omega$$

This can be verified in the lab (Part 1.5) by inserting a series resistor R_s = 10 kΩ between the source and R_1, observing the attenuation, and applying the voltage divider relationship.

Phase Relationship

The inverting amplifier produces a 180° phase shift between input and output. For an AC input, the output will be an inverted replica of the input, scaled by the gain magnitude of

Part 2 – Non-Inverting Amplifier

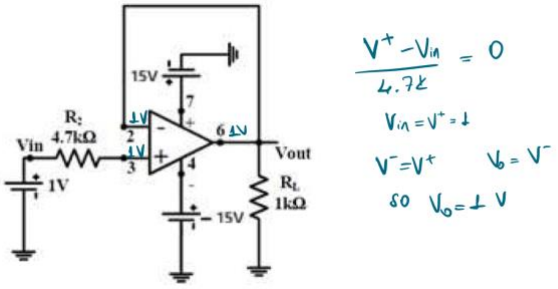


Figure 4

Circuit Description

The non-inverting amplifier in Figure 4 consists of:

- R_2 = 4.7 kΩ (from inverting input to ground)
- R_L = 1 kΩ (load resistor at output)
- Feedback wire connects output directly to inverting input through the resistor network
- V_in = 1 V DC applied to non-inverting input (+), ±15 V supply

Transfer Function Derivation

For the general non-inverting amplifier with feedback resistors R_2 (from inverting input to ground) and a feedback path:

Using the virtual short condition: $V^- = V^+ = V_in$

The voltage at the inverting input is set by the voltage divider between R_L (or output) and R_2:

$$V^- = R_2 / (R_2 + R_feedback) \times V_out$$

Since V^- = V_{in}:

$$V_in = R_2 / (R_2 + R_feedback) \times V_out$$

$$A_v = V_out / V_in = 1 + R_feedback / R_2$$

For Figure 4 specifically, with a direct wire feedback (R_feedback = 0):

$$A_v = 1 + 0 / R_2 = 1$$ so **V_out = V_in = 1 V**

Currents Through Resistors (Expected Values)

I_4.7k = V^- / R_2 = 1 V / 4.7 kΩ ≈ 0.213 mA

I_1k = V_out / R_L = 1 V / 1 kΩ = 1 mA

I_feedback ≈ 0 (ideal op-amp: no current into input terminal)

Virtual Ground Condition

In a non-inverting amplifier, the virtual ground condition does NOT mean pin 2 is at 0 V. Instead, pin 2 equals pin 3 (both equal V_in). Therefore, there is no virtual ground in the traditional sense – instead there is a "virtual short" between the two input terminals.

V_pin3 = V_pin2 = 1 V

This is the key difference from the inverting amplifier, where pin 2 is held at virtual ground (0 V).

3. Data/Results:

Building Block: Part 1 Inverting Amplifier

This part of the experiment investigates the behavior of an inverting operational amplifier configuration. The circuit was built using an LM741 op-amp with a feedback resistor network to control the voltage gain. The input signal is applied to the inverting terminal of the op-amp through resistor R_1, while the feedback resistor R_F connects the output back to the same node. The non-inverting terminal is connected to ground through a resistor to maintain stability. Due to the high open-loop gain of the op-amp and the negative feedback configuration, the voltage at the inverting input behaves as a virtual ground. In this configuration, the theoretical voltage gain of the amplifier is determined by the ratio of the feedback resistor to the input resistor.

Components

LM741 operational amplifier
5.6 kΩ resistor (R1)
56 kΩ resistor (RF)
4.7 kΩ resistor (R2)
DC power supply (±15 V)
Function generator
Oscilloscope
Breadboard and connecting wires

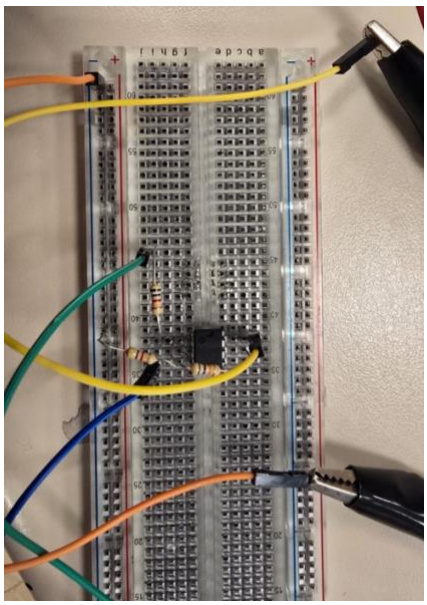


Figure 2 – Breadboard implementation of the inverting amplifier circuit.

Description

The inverting amplifier circuit shown in Figure 1 was constructed on a breadboard using an LM741 operational amplifier. The circuit was powered using ±15 V supply voltages. The input voltage was initially set to 1 V DC in order to analyze the current distribution in the circuit.

Since the inverting input of the operational amplifier behaves as a virtual ground, the current through the input resistor R_1 flows entirely through the feedback resistor R_F. The measured currents through the resistors were used to verify this behavior and confirm the virtual ground condition at pin 2.

Measurement

Part 1.1 – Current Measurements

Measured currents in the circuit: IR1 = 0.176 mA, IR2 = 0 mA, IRF = 0.179 mA

Measured currents in the circuit: IR1 = 0.176 mA, IR2 = 0 mA, IRF = 0.179 mA. The currents were measured by placing the multimeter in series with the corresponding resistor in the circuit. The measured values of I_{R1} and I_{RF} are approximately equal, which confirms that the inverting input node behaves as a virtual ground and that almost no current flows into the op-amp input terminal. The current through R_2 was measured to be approximately zero, supporting the assumption that the node is effectively at ground potential.

Part 1.2 – Voltage Gain Calculation

The voltage gain of an inverting amplifier is theoretically given by Vo / Vin = −RF / R1. Using the resistor values in the circuit RF = 56 kΩ, R1 = 5.6 kΩ the theoretical gain becomes Vo / Vin = −56k / 5.6k = −10

This indicates that the output signal should be inverted and amplified by a factor of 10 relative to the input.

Part 1.3 – Measured Voltage Gain

The output voltage was measured using a voltmeter connected to pin 6 of the operational amplifier and ground. The measured voltage gain of the circuit was

Vo / Vin = −9.93

This value is very close to the theoretical gain of −10. The small difference can be attributed to resistor tolerances, measurement inaccuracies, and the non-ideal characteristics of the operational amplifier.

Part 1.4 – AC Signal Measurement

In this step, the DC input source was replaced with a function generator. The function generator was configured to produce a sinusoidal signal with an amplitude of 500 mV peak-to-peak and a frequency of 1 kHz. Channel 1 of the oscilloscope was connected to the input signal while channel 2 was connected to the output of the operational amplifier.

The oscilloscope measurements showed that the input signal had a peak-to-peak value of approximately 510 mV, while the output signal had a peak-to-peak value of approximately 5.1 V. Based on these measurements, the voltage gain of the amplifier was calculated and measured as

Vo / Vin ≈ 5.1 V / 0.51 V ≈ 10

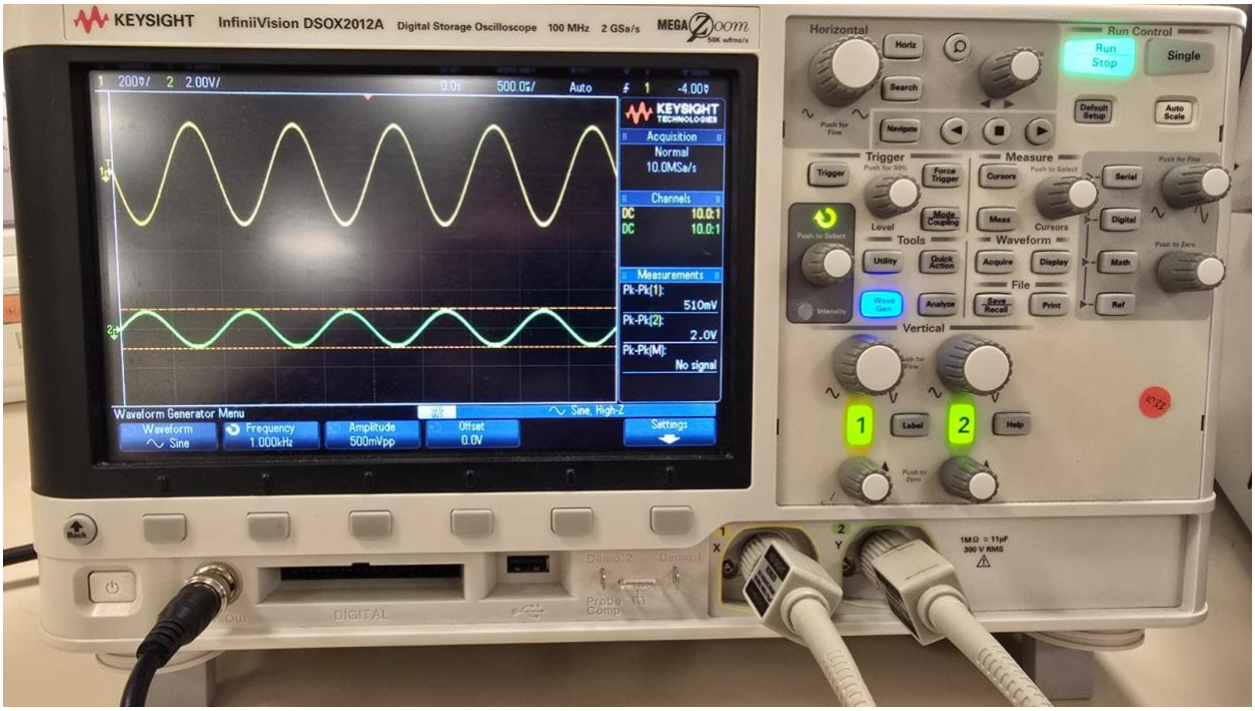


Figure 3 – Oscilloscope display showing the input signal (Channel 1) and the amplified, inverted output signal (Channel 2) of the inverting amplifier.

The oscilloscope display also showed that the output signal is inverted with respect to the input signal. This corresponds to a phase shift of approximately 180°, which is the expected behavior for an inverting amplifier configuration. The measured gain is very close to the theoretical gain value of −10, confirming the correct operation of the circuit.

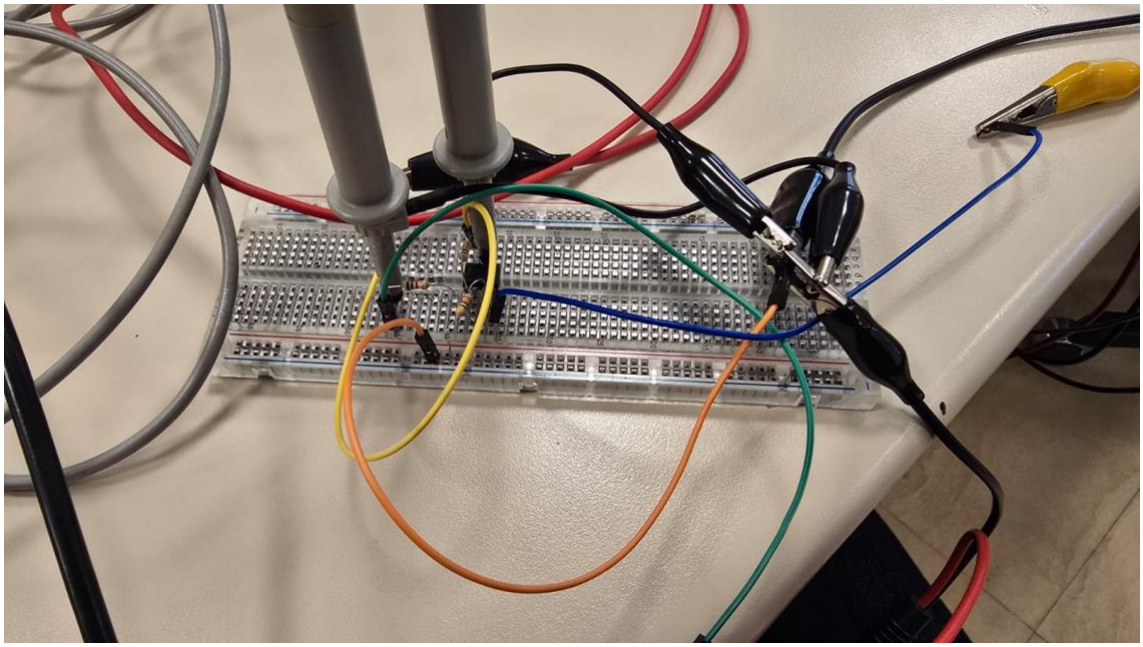


Figure 5 – Breadboard implementation of the inverting amplifier circuit used for AC signal measurements.

Part 1.5 – Input Impedance Measurement

In this step, a 10 kΩ resistor was connected between the function generator and the input resistor R_1 as shown in Figure 6. This additional resistor forms a voltage divider with the input impedance of the inverting amplifier. By observing the change in voltage gain on the oscilloscope (Figure 7), the input impedance of the amplifier can be estimated.

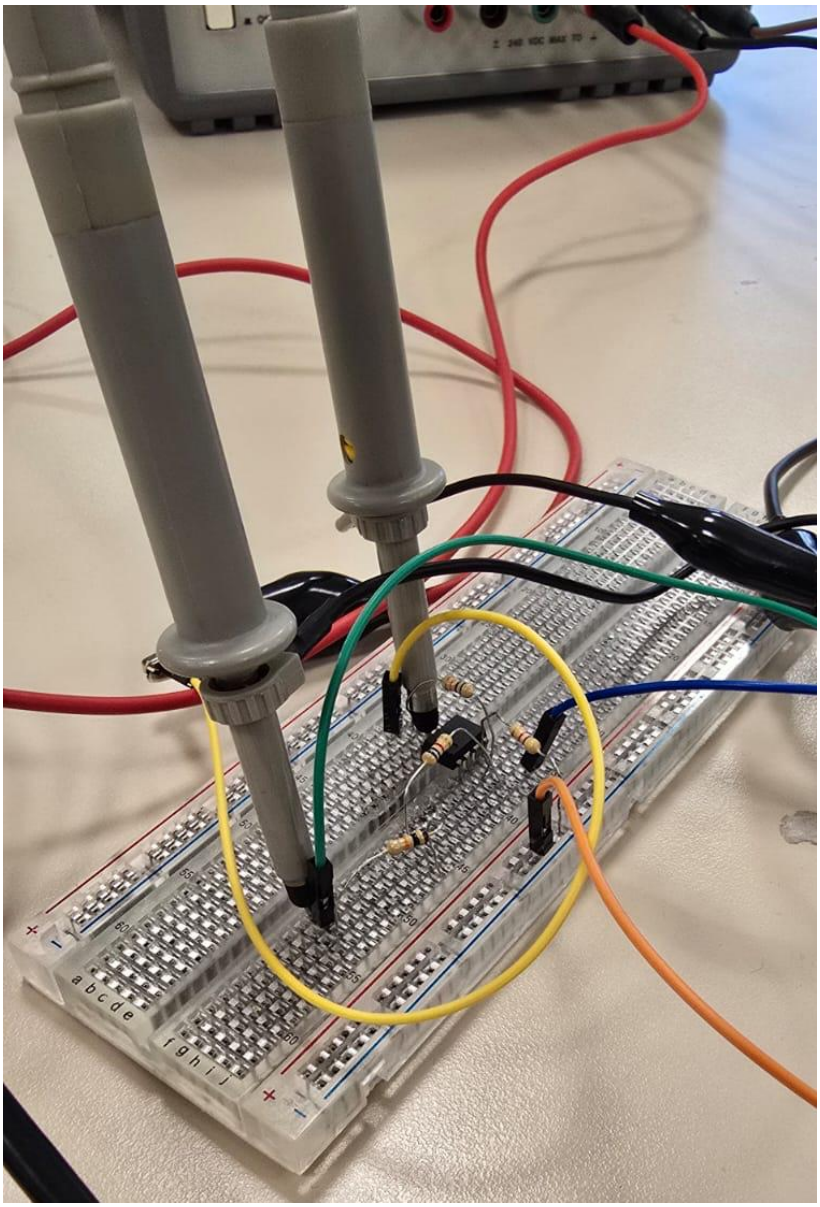


Figure 6 – Breadboard implementation of the circuit after inserting the 10 kΩ resistor in series with the input to measure the input impedance.

After inserting the 10 kΩ resistor, the measured output voltage was approximately 1.9 V for an input signal of approximately 0.51 V. Therefore, the new voltage gain was calculated and measured as

$$V_o/V_{in} = 1.9\ \text{V}/0.51\ \text{V} \approx 3.73$$

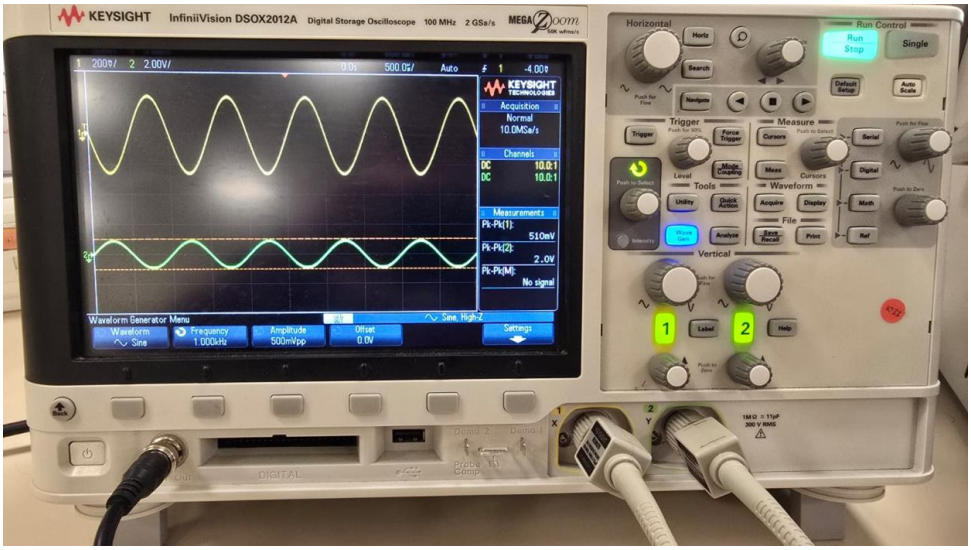


Figure 7 – Oscilloscope display showing the reduced output amplitude due to the voltage divider formed by the added 10 kΩ resistor and the amplifier input impedance.

Using the voltage divider relationship, the input impedance of the amplifier can be determined. The total resistance seen by the source becomes the sum of the added 10 kΩ resistor and the input resistor of the amplifier. Therefore, the input impedance was estimated as

$$R_{in} \approx 5.6\ k\Omega + 10\ k\Omega \approx 15.6\ k\Omega$$

This result shows that the input impedance of the inverting amplifier is approximately equal to the input resistor R_1, which is consistent with the theoretical behavior of an inverting op-amp configuration.

Part 1.6 – Rise Time, Fall Time and Slew Rate Measurement

In this part, the input waveform was changed from a sine wave to a square wave with an amplitude of 1 V and a frequency of 10 kHz. Due to the limited slew rate of the op-amp, the output signal was not an ideal square wave. Instead, the transitions between voltage levels appeared as sloped edges rather than instantaneous vertical steps.

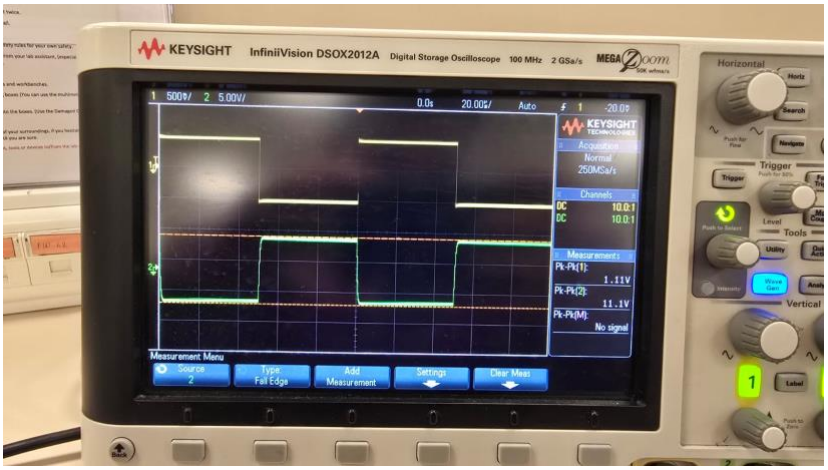


Figure 8 – Oscilloscope showing the square wave input (yellow) and the inverted output (green) with finite rise and fall slopes.

To measure the rise and fall times, the oscilloscope cursor function was used. The cursors were placed on the waveform between the 10% and 90% voltage levels of the transition. The horizontal cursor distance (ΔX) provided the corresponding transition time directly.



Figure 9 – Oscilloscope screenshots showing the output transitions used to measure the rise time and fall time of the op-amp signal.

From the measurements, the rise time was approximately 1.1 µs and the fall time was approximately 0.98 µs. The voltage change between the 10% and 90% levels was approximately 8.8125 V. Using the relation Slew Rate = ΔV / T, the rising slew rate was calculated as approximately 8.01 V/µs and the falling slew rate as approximately 8.99 V/µs.

The obtained waveform clearly shows the finite slope during transitions, which indicates the slew rate limitation of the op-amp. The slew rate represents the maximum rate at which the output voltage of the amplifier can change with respect to time.

Building Block: Part 2 Non-Inverting Amplifier

In this part, a non-inverting amplifier configuration using an LM741 operational amplifier was implemented on the breadboard. The circuit was powered with ±15 V supply voltages to ensure proper operation of the op-amp.

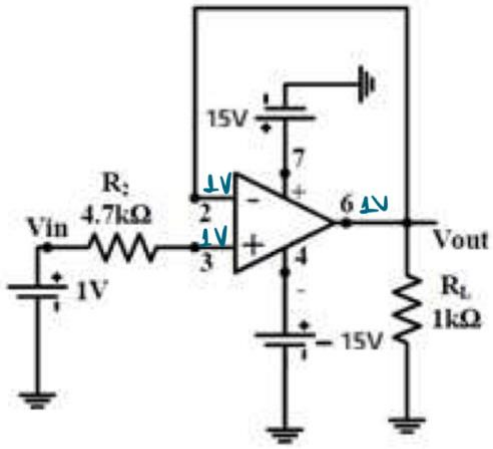


Figure 4

The input voltage $V_{in} = 1$ V was applied to the non-inverting input (pin 3) through a 4.7 kΩ resistor. The output of the op-amp (pin 6) was directly connected to the inverting input (pin 2), creating a unity-gain feedback configuration. A 1 kΩ load resistor R_L was connected between the output and ground.

The main components used in this circuit were:
LM741 operational amplifier
4.7 kΩ input resistor
1 kΩ load resistor
±15 V DC power supplies
Breadboard and connecting wires
Oscilloscope probes for measurement

This configuration operates as a voltage follower (buffer), where the output voltage follows the input voltage. Due to the negative feedback, the op-amp forces the voltages at the inverting and non-inverting inputs to be approximately equal. Therefore, the output voltage becomes approximately equal to the input voltage while providing the ability to drive the load resistor without loading the input source.

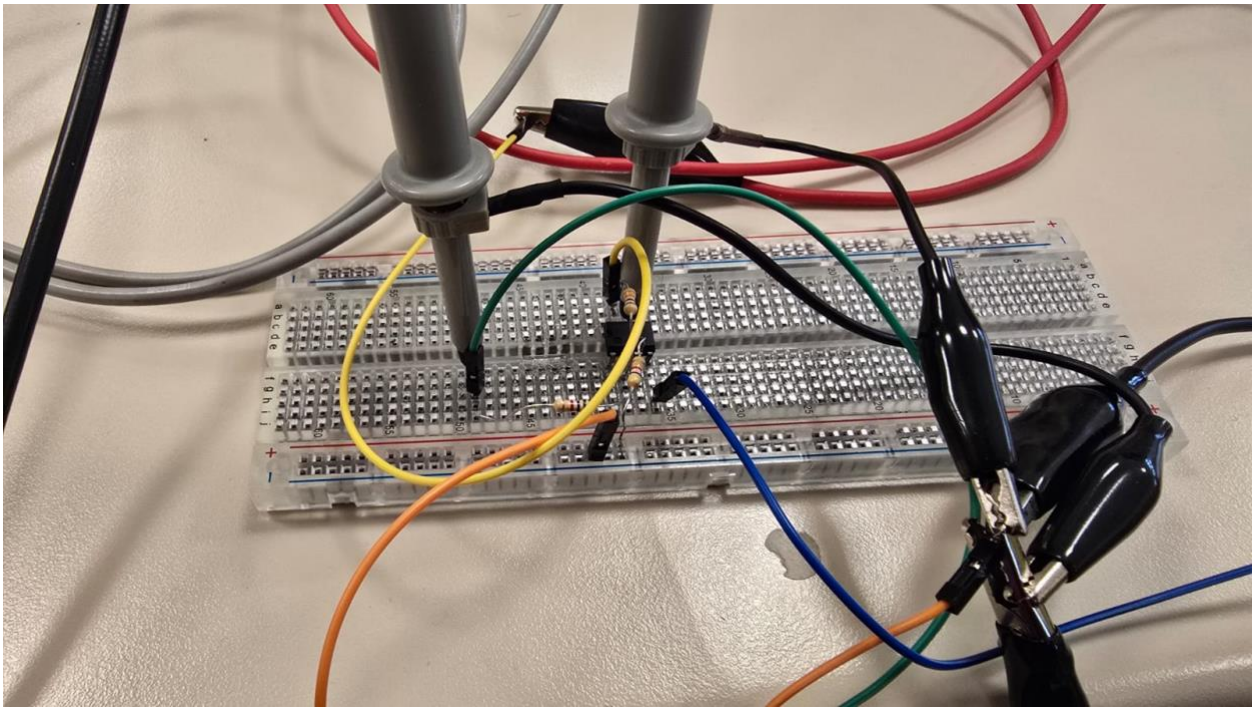


Figure 10. Breadboard implementation of the non-inverting amplifier circuit with the oscilloscope probes connected to measure the input and output voltages.

Part 2.1 – Current Measurement

In this step, the circuit shown in the figure was built on the breadboard using an LM741 operational amplifier powered with ±15 V supplies. The input voltage was set to $V_{in} = 1$ V. Since the output is directly fed back to the inverting input, the circuit operates as a unity-gain buffer, meaning the output voltage is approximately equal to the input voltage.

To measure the currents through the resistors and the feedback path, the current was measured **in series** with each element. The wire was temporarily opened and the ammeter was inserted in series so that the current flowing through the component passed through the measuring device.

First, the current through the **4.7 kΩ input resistor** was measured. Because the input of the op-amp ideally draws almost no current, the measured current was approximately **0 A**.

Next, the current through the **1 kΩ load resistor** was measured. Since the output voltage was approximately $V_{out} \approx 1$ V, the measured current was about **0.98 mA**, which is consistent with Ohm's law:

$$I = \frac{V}{R} = \frac{1\ \text{V}}{1\ \text{k}\Omega} \approx 1\ \text{mA}$$

Finally, the **feedback current** was measured. Because the feedback path is directly connected to the high-impedance op-amp input, the measured current in the feedback wire was approximately **0 A**.

These results confirm the theoretical behavior of a non-inverting voltage follower, where the op-amp input draws negligible current while the output supplies the current required by the load resistor.

Part 2.2 – DC Voltage Measurement

In this step, the DC voltages at pin 2 (inverting input) and pin 3 (non-inverting input) of the operational amplifier were measured with respect to ground using a multimeter. The purpose of this measurement was to verify whether the input terminals of the op-amp operate at the same potential due to the negative feedback.

The measured voltage **at pin 3 was approximately 1 V**, which is equal to the applied input voltage V_{in}. The voltage **at pin 2 was also measured to be approximately 1 V.** This occurs because the op-amp, operating with negative feedback, forces the voltages at its two input terminals to become nearly equal.

Therefore, the condition $V^- \approx V^+$ is satisfied. However, unlike the inverting amplifier configuration, this voltage is not equal to ground, but instead equal to the input voltage. For this reason, the virtual ground condition does not exist in the non-inverting amplifier configuration. Instead, both inputs follow the input voltage level while remaining nearly equal to each other due to the high gain and feedback of the op-amp.

Part 2.3 – DC Output Voltage and Voltage Gain

In this step, the DC voltage across the **1 kΩ load resistor** was measured in order to determine the output voltage and calculate the voltage gain of the circuit. The voltage was measured using a multimeter by placing the probes across the load resistor terminals.

The measured output voltage was approximately $V_o = 1$ V. Since the input voltage was also $V_{in} = 1$ V, the voltage gain of the circuit was calculated as

$$\frac{V_o}{V_{in}} = \frac{1\ \text{V}}{1\ \text{V}} = 1$$

This confirms the expected behavior of the **non-inverting voltage follower**, where the output voltage follows the input voltage with a gain of approximately one.

While performing the measurement, the multimeter probes had to be held on both terminals of the resistor simultaneously, which required using both hands. Therefore, I cannot take a photograph of the measurement during the process.

Part 2.4 – AC Input Signal and Voltage Gain

In this step, the 1 V DC source was replaced with a function generator. The function generator was configured to produce a 1 V peak sine wave at a frequency of 1 kHz. The input signal was measured using Channel 1 of the oscilloscope, while the output signal was measured using Channel 2.

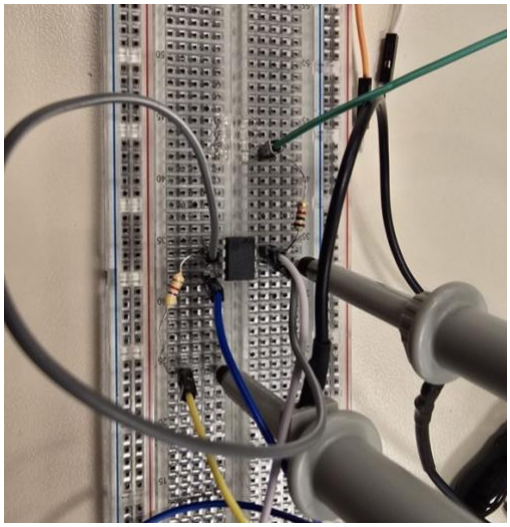


Figure 11 – Breadboard implementation of the non-inverting amplifier circuit with oscilloscope probes connected to measure the input and output signals.

As shown in the oscilloscope measurement, the peak-to-peak voltages of both signals were approximately 2.2 Vpp. Therefore, the voltage gain of the circuit was calculated as

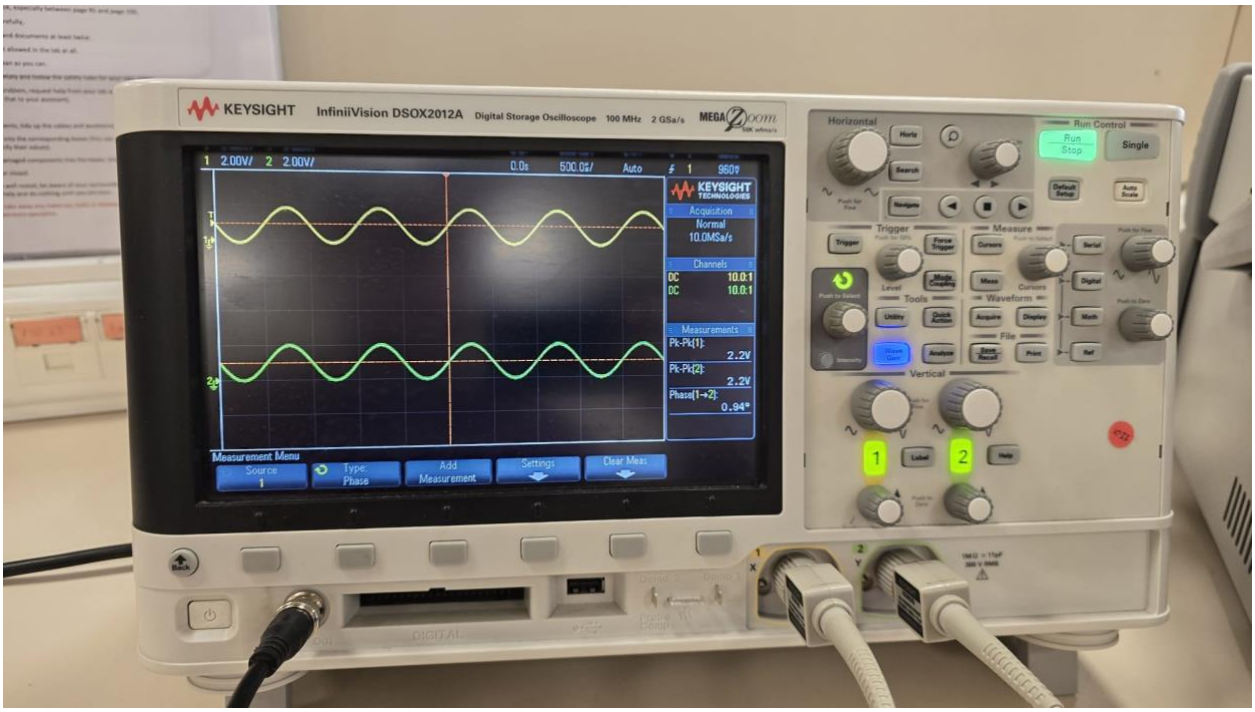


Figure 12 – Oscilloscope display showing the input (Channel 1) and output (Channel 2) sine waves with approximately equal amplitudes and nearly zero phase difference.

$$\frac{V_o}{V_{in}} = \frac{2.2\ \text{V}}{2.2\ \text{V}} \approx 1$$

This result confirms that the non-inverting amplifier in this configuration behaves as a voltage follower, where the output amplitude closely follows the input amplitude.

The oscilloscope also displayed a phase difference of approximately 0.94°, which is very close to zero. This indicates that the input and output signals are nearly in phase, which is expected for a non-inverting amplifier configuration.

Part 2.5 – Input Impedance of the Non-Inverting Amplifier

The same technique used in Part 1.5 cannot be effectively applied to measure the input impedance of the non-inverting amplifier. In the inverting amplifier, the input impedance is approximately equal to the input resistor R_1, so adding a series resistor creates a noticeable voltage divider that allows the input impedance to be estimated.

However, in the non-inverting amplifier configuration, the input signal is applied directly to the non-inverting input of the op-amp, which ideally has an extremely high input impedance (ideally infinite). Because of this very high impedance, almost no current flows into the input terminal.

If a resistor were added in series with the input signal, it would form a voltage divider with the op-amp input impedance. Since the op-amp input impedance is extremely large compared to the added resistor, the voltage drop across the added resistor would be negligible. As a result, the input voltage would remain nearly unchanged and the voltage gain would not noticeably change.

Therefore, this method cannot reliably be used to determine the input impedance of the non-inverting amplifier. In practice, the input impedance of this configuration is very high and dominated by the intrinsic input impedance of the operational amplifier itself, rather than by an external resistor as in the inverting amplifier.

4. Discussion/Analysis

The results obtained in this experiment were largely consistent with the theoretical behavior of operational amplifiers operating with negative feedback. The main objectives of the experiment were to investigate the characteristics of inverting and non-inverting amplifiers, verify theoretical gain relationships, estimate input impedance, and observe practical limitations of real operational amplifiers such as slew rate. The measured results summarized in the Results section generally matched the expected theoretical values.

For the inverting amplifier, the theoretical voltage gain is given by the relationship $A_v = -\frac{R_f}{R_1}$
Using $R_f = 56\ k\Omega$ and $R_1 = 5.6\ k\Omega$, the expected gain is −10. The measured gain was approximately −9.93, which is very close to the theoretical prediction. This small difference represents a minor deviation from the ideal value and can be explained by resistor tolerances, measurement uncertainty in the oscilloscope readings, and the finite open-loop gain of the LM741 operational amplifier. In practice,

operational amplifiers do not behave as ideal devices, so small discrepancies between theoretical and measured values are expected.

The current measurements further confirmed the theoretical assumptions of the ideal op-amp model. The measured currents through the input and feedback resistors were approximately $I_{R1} = 0.176\,mA$ and $I_{RF} = 0.179\,mA$, while the current entering the op-amp input was approximately zero. Since the currents through R_1 and R_f are nearly equal, this indicates that almost no current flows into the op-amp input terminal. This observation supports the assumption of extremely high input impedance and verifies the concept of **virtual ground**, where the inverting input node remains very close to 0 V even though it is not physically connected to ground.

In the input impedance experiment, inserting a 10 kΩ resistor in series with the signal source caused the measured gain to decrease to approximately 3.73. This behavior is expected because the added resistor forms a voltage divider with the input resistance of the amplifier. As a result, the signal reaching the amplifier input is reduced. From this behavior, the input impedance was estimated to be approximately 15.6 kΩ. Although this value differs somewhat from the theoretical value based solely on the input resistor, such differences may arise from oscilloscope reading precision, resistor tolerances, and small parasitic resistances in the breadboard connections.

Another important result of this experiment was the observation of the **slew rate limitation** of the op-amp. When the input signal was changed to a square wave at 10 kHz, the output waveform did not appear as a perfect square wave. Instead, the transitions between voltage levels occurred gradually, indicating that the op-amp output cannot change instantaneously. Using oscilloscope cursors, the rise time was measured to be approximately 1.1 µs and the fall time approximately 0.98 µs. With a measured voltage change of about 8.8125 V, the corresponding slew rates were calculated as approximately 8.01 V/µs for the rising edge and 8.99 V/µs for the falling edge. These values demonstrate the finite speed of the op-amp and illustrate a key non-ideal limitation of practical amplifier circuits.

For the non-inverting amplifier configuration, the circuit behaved as a voltage follower. The measured DC gain was approximately 1, with both the input and output voltages measured around 1 V. The AC measurements also confirmed this behavior. When a 1 V peak sine wave at 1 kHz was applied, both the input and output signals were measured to be approximately 2.2 V peak-to-peak, and the phase difference between them was about 0.94°, indicating that the signals were nearly in phase. These results are consistent with the theoretical behavior of a non-inverting amplifier with unity gain.

The measured currents in the non-inverting amplifier further support the theoretical model. Almost no current flowed through the input resistor, while the load current was approximately 0.98 mA, which is consistent with Ohm's law for the 1 kΩ load resistor. This confirms that the non-inverting input of the op-amp has extremely high input impedance and that the output stage of the amplifier supplies the current required by the load.

Finally, the method used earlier to estimate the input impedance of the inverting amplifier could not be applied effectively to the non-inverting amplifier. In the inverting configuration, the input impedance is determined mainly by the input resistor, allowing a measurable voltage divider to be formed. In contrast, the non-inverting amplifier presents a very large input impedance because the signal is applied directly to the op-amp input. As a result, almost no current flows into the input terminal, and inserting a series resistor does not significantly change the measured gain.

Overall, the experimental observations were in strong agreement with theoretical predictions. Small deviations from ideal values are expected due to component tolerances, oscilloscope measurement resolution, and the non-ideal characteristics of the LM741 operational amplifier. Additional differences may also arise from breadboard parasitic resistances and capacitances, as well as wiring effects in the experimental setup. These results demonstrate both the accuracy of op-amp circuit theory and the practical limitations of real electronic components.

Summary of Results

Table 1 – Inverting Amplifier Results

Quantity	Theoretical Value	Measured Value	Comments
Voltage Gain	-10	-9.93	Very close to theory, small error due to resistor tolerance
Input Current (Op-Amp)	≈ 0 A	≈ 0 A	Confirms high input impedance
Current through R1	≈ 0.179 mA	0.176 mA	Matches expected current flow
Current through Rf	≈ 0.179 mA	0.179 mA	Nearly identical to input current
Phase Shift	180°	≈180°	Confirms inverting behavior
Estimated Input Impedance	5.6 kΩ	≈15.6 kΩ	Includes added 10 kΩ series resistor
Rising Slew Rate	—	8.01 V/μs	Limited by op-amp internal characteristics
Falling Slew Rate	—	8.99 V/μs	Slight difference due to measurement error

Table 2 – Non-Inverting Amplifier Results

Quantity	Theoretical Value	Measured Value	Comments
Voltage Gain (DC)	1	1	Voltage follower behavior
Voltage Gain (AC)	1	≈1	Input and output amplitudes equal
Phase Difference	0°	0.94°	Nearly in phase
Current through 4.7 kΩ	≈0 A	≈0 A	Confirms very high input impedance
Current through Load (1 kΩ)	1 mA	0.98 mA	Matches Ohm's law
Pin 2 Voltage	1 V	1 V	$V^- \approx V^+$ due to feedback
Pin 3 Voltage	1 V	1 V	Input voltage

6. Conclusion

The results of this experiment generally agree with the theoretical behavior of operational amplifiers with negative feedback. In the inverting amplifier configuration, the theoretical gain was −10, while the measured gain was approximately −9.93, which shows very good agreement with theory. The measured resistor currents also confirmed the ideal op-amp assumption that almost no current flows into the input terminals and that the inverting input behaves as a virtual ground.

In the non-inverting configuration, the circuit operated as a voltage follower. The measured output voltage was approximately equal to the input voltage and the phase difference between input and output was nearly zero, confirming the expected unity gain and in-phase behavior predicted by theory.

The slew rate experiment demonstrated a key limitation of real operational amplifiers. When a high-frequency square wave was applied, the output transitions were not instantaneous, producing measured slew rates of about 8–9 V/µs. This shows that practical op-amps cannot change their output voltage infinitely fast.

The experimental results validate the theoretical models of inverting and non-inverting amplifiers. Small differences between theoretical and measured values can be explained by resistor tolerances, non-ideal op-amp characteristics, parasitic effects of the breadboard, and measurement uncertainties. Future experiments using more precise components and measurements at different frequencies could further analyze these non-ideal effects.